

ANDRII VYNOHRADOV

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Work Authorization: Canada Open Work Permit (Legally Authorized to Work for Any Canadian Employer - No Visa Sponsorship Required)

Portfolio: vynohradov.ca

PROFESSIONAL SUMMARY

Systems Architect & Lead Backend Engineer with 7+ years of commercial experience building high-throughput distributed systems, real-time C++ applications, and production AI platforms. Specialized in resilient backend architectures (Node.js/NestJS, Python), local model inference (vLLM, Ollama), and low-latency hardware/audio communication protocols. Track record includes optimizing p95 orchestration latencies, eliminating 70% of cloud storage costs via bare-metal S3 deployments, and driving technical standards (RFC/ADR) across engineering teams.

TECHNICAL SKILLS

- **AI & LLM Systems:** Agent Orchestration, Hybrid RAG, Vector Search, Local Model Inference (vLLM, Ollama), Model Quantization (GGUF, AWQ), LangGraph, Vercel AI SDK, Tool Calling Pipelines
- **Systems & Low-Latency:** C++20, JUCE Framework, Lock-Free Concurrency, Zero-Allocation Real-Time Systems, Inter-Process Communication (IPC Sockets), DSP, Hardware Protocols (DMX-512, Art-Net, sACN, MIDI, ESP32 UDP)
- **Backend & Cloud Architecture:** Node.js, NestJS, Python (FastAPI, Django), C# (.NET Core / ASP.NET), Microservices, Event-Driven Architecture, Redis, BullMQ, PostgreSQL (JSONB, Query Tuning), gRPC, WebSockets, REST APIs, GraphQL
- **Infrastructure, DevOps & Linux:** Linux Administration (Ubuntu Bare-Metal), Docker, Kubernetes, Traefik, MinIO (S3 Object Storage), CI/CD (GitHub Actions, GitLab CI), GPU Acceleration (CUDA), WireGuard Mesh VPN
- **Frontend & Client Platforms:** TypeScript, React, Next.js, React Native, Redux Toolkit, Zustand, Tailwind CSS, Three.js (WebGL), Vite, Electron

PROFESSIONAL EXPERIENCE

IndagoDev LLC | Remote

Lead AI Platform Engineer & Systems Architect | Jun 2024 - Present

- Decomposed a legacy Django monolith into an event-driven Node.js / NestJS microservices platform, resolving concurrency bottlenecks to support 1,000+ parallel AI agents while cutting p95 orchestration gateway latency from 450ms to 315ms under production load.
- Engineered a production Hybrid RAG pipeline combining vector similarity search with structured metadata pre-filtering; built a dual-tier agent memory store that reduced retrieval noise and cut prompt token overhead by ~40%.
- Built an extensible execution engine to orchestrate multi-step agent workflows across LLMs, code sandboxes, and web scrapers with type-safe streaming and automated retry handling.
- Cut monthly cloud infrastructure expenses by 70% by deploying a distributed MinIO S3-compatible storage cluster on bare-metal Linux servers, eliminating third-party cloud data egress fees.
- Provisioned isolated multi-tenant container/VM runtime environments for autonomous agents, managing 100+ high-utilization instances with automated health checks and self-healing recovery workflows.
- Architected real-time WebSocket telemetry streaming and client integration, delivering sub-100ms end-to-end agent execution updates for 500+ active beta users.
- Authored 12+ RFCs and ADRs to streamline architectural decisions, mentored 5 engineers, and established team-wide backend engineering standards.

b4value.net GmbH | Grünstadt, Germany

Software Developer | May 2022 - Jun 2024

- Designed modular React/TypeScript frontend micro-architectures for compliance-regulated enterprise document management platforms handling sensitive legal and financial transactions.
- Modernized frontend build tooling by migrating legacy Webpack configurations to Vite and Rollup, cutting CI/CD build times from 12m to 4.2m across core enterprise repositories.
- Built high-throughput ASP.NET Core / C# API endpoints, optimizing bidirectional data transfer pipelines and client-side caching strategies using Redux Toolkit.
- Refactored 4 core document processing applications to achieve 100% compliance with WCAG 2.1 and German BITV accessibility standards, securing enterprise client renewal validations.

Uk. Soft - Evolution | Dnipro, Ukraine

Full-Stack Software Developer | Sep 2018 - May 2022

- Developed and maintained custom e-commerce platforms, workflow automation services, and REST APIs using PHP and MySQL.
- Designed normalized relational database schemas, implemented secure payment gateway integrations, and managed Linux server deployments.
- Handled client requirements discovery, sprint planning, and end-to-end application delivery across the full software lifecycle.

ENGINEERING PROJECTS

ResoStage - Real-Time Live DAW & Lighting Sequencer

Tech Stack: C++20, JUCE 9, Lock-Free Concurrency, Electron, React 19, Three.js, WebSockets, DMX-512 / Art-Net / sACN

- Engineered a cross-platform live audio workstation operating with zero dynamic heap allocations in the real-time audio thread, utilizing precomputed Sinc interpolation for sub-millisecond pitch shifting and sample-accurate clock sync.
- Implemented lock-free atomic queues and streaming ring buffers with elevated thread I/O scheduling, eliminating audio dropouts during heavy disk read stalls.
- Built a native process supervisor daemon (Kaishaku) operating over IPC heartbeat channels to handle instant process isolation and state-preserving hot restarts upon UI crashes.
- Architected a multi-protocol 60 Hz lighting engine driving DMX-512, Art-Net, sACN (E1.31), and custom ESP32 UDP protocols, paired with a Three.js WebGL 3D real-time stage visualizer.

Self-Hosted Private Cloud & Local AI Inference Infrastructure

Tech Stack: Bare-Metal Ubuntu, Docker, Traefik, MinIO, vLLM, Ollama, WireGuard, systemd

- Deployed containerized GPU inference pipelines using vLLM and Ollama for running open-source LLMs locally with optimized throughput and VRAM allocation.
- Configured secure cross-node mesh networking using WireGuard and Traefik reverse proxy with automated Let's Encrypt TLS renewal.
- Built a clustered MinIO S3 object storage backend with automated encrypted offsite backups.

VST/AU Plugin Suite

Tech Stack: C++, JUCE Framework, Digital Signal Processing (DSP), AI-Assisted Development.

- Engineered a proprietary suite of VST/AU plugins optimized for live performances, including a dual-deck scratch emulator ("Scratcher") and a modernized software synthesizer ("Floppy Synth").
- Implemented complex digital signal processing algorithms using C++ and the JUCE framework, with a strict focus on real-time thread safety, memory management, and near-zero latency within professional DAWs.
- Leveraged AI tools to rapidly scaffold boilerplate JUCE components, optimizing development cycles to focus on critical DSP logic and architectural stability.

EDUCATION

Vocational Diploma: Fachinformatiker für Anwendungsentwicklung

IHK-Certified Dual Vocational Program | Germany

- Completed Germany's recognized dual-track software engineering curriculum ahead of regular schedule with honors.
- Credential Recognition: Evaluated by WES as equivalent to a Canadian Applied Computer Science Diploma.

Undergraduate Coursework in Computer Science & Engineering

Prydniprovsk State Academy of Civil Engineering and Architecture | Ukraine

- Completed 2 years of full-time coursework focusing on Data Structures, Algorithms, and Software Engineering prior to relocating to Germany.